

Introduction to Machine Learning

Lecture 1

Machine Learning

- Machine Learning is the science of getting computers to learn without being explicitly programmed.”
- Computers usually need step-by-step rules
- ML learns rules from data

Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

- improve their performance P
- at some task T
- with experience E .

A well-defined learning task is given by $\langle P, T, E \rangle$.

T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

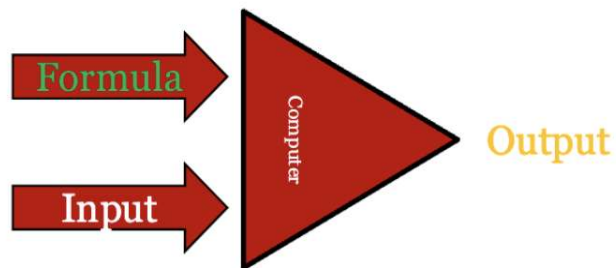
T: Recognizing hand-written words

P: Percentage of words correctly classified

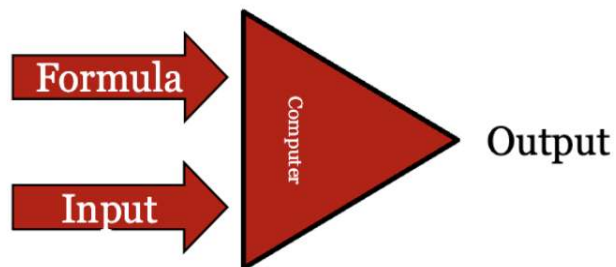
E: Database of human-labeled images of handwritten words

ML vs Traditional sw development

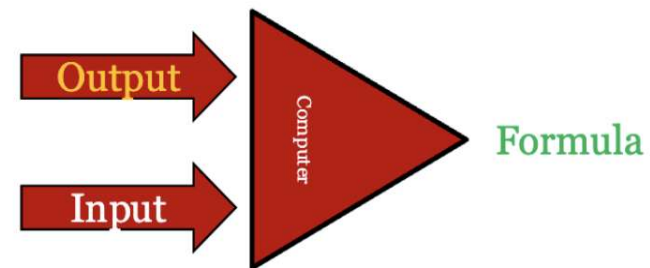
Traditional Software during development



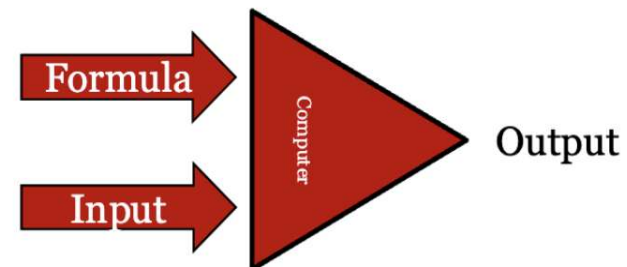
Traditional Software during deployment



Machine Learning during development/Training



Machine Learning during deployment/Inference

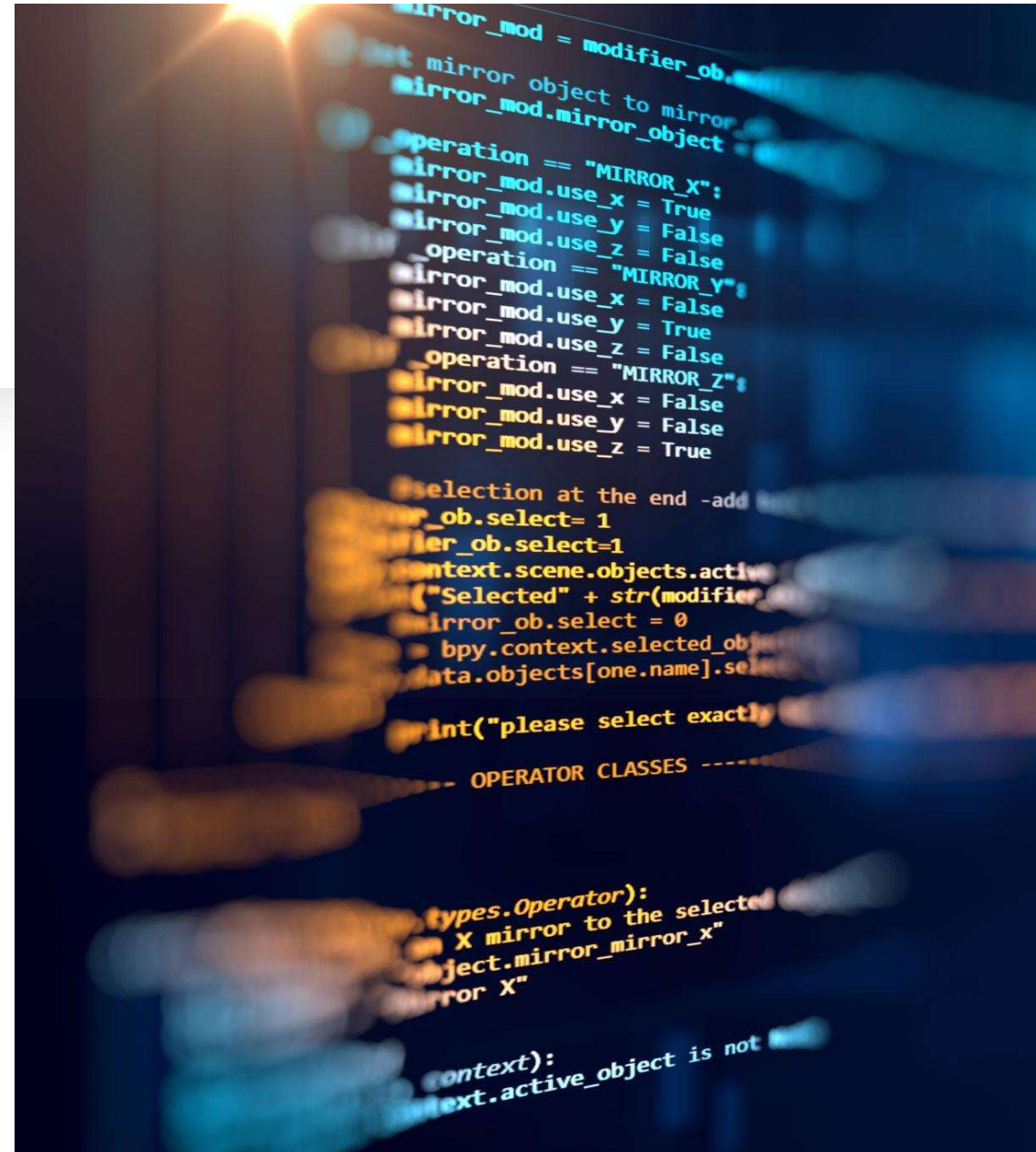


Why ML matters

- ML is already everywhere.
- Netflix / Spotify recommendations
- Google Maps travel time prediction
- Spam filtering
- Self-driving cars
- Face unlock
- Industrial automation

Why Now

- 3 reasons ML exploded:
- **More data**
- **Faster computers**
- **Better algorithms**



Some examples

- Recognizing patterns:
 - Facial identities or facial expressions
 - Handwritten or spoken words
 - Medical images
- Generating patterns:
 - Generating images or motion sequences
- Recognizing anomalies:
 - Unusual credit card transactions
 - Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
 - Future stock prices or currency exchange rates

When Should We Use Machine Learning?

- ML is a great fit when...
- **Scale:**
The problem is too big or too fast for humans to handle manually
- **Change:**
The patterns in data keep changing and rules need constant updating
- **Complexity:**
The logic is too complicated to write as simple rules — ML can learn it from data

ML should NOT be used when...

- **Not enough data / poor data quality**
(You can't learn patterns from bad or tiny datasets)
- **A simple rule works fine**
("If temperature > 30°C → turn on AC" — no ML needed)
- **It is not cost-effective**
ML takes time, money, and maintenance — use it only when the benefit outweighs the cost

Supervised vs Unsupervised Learning

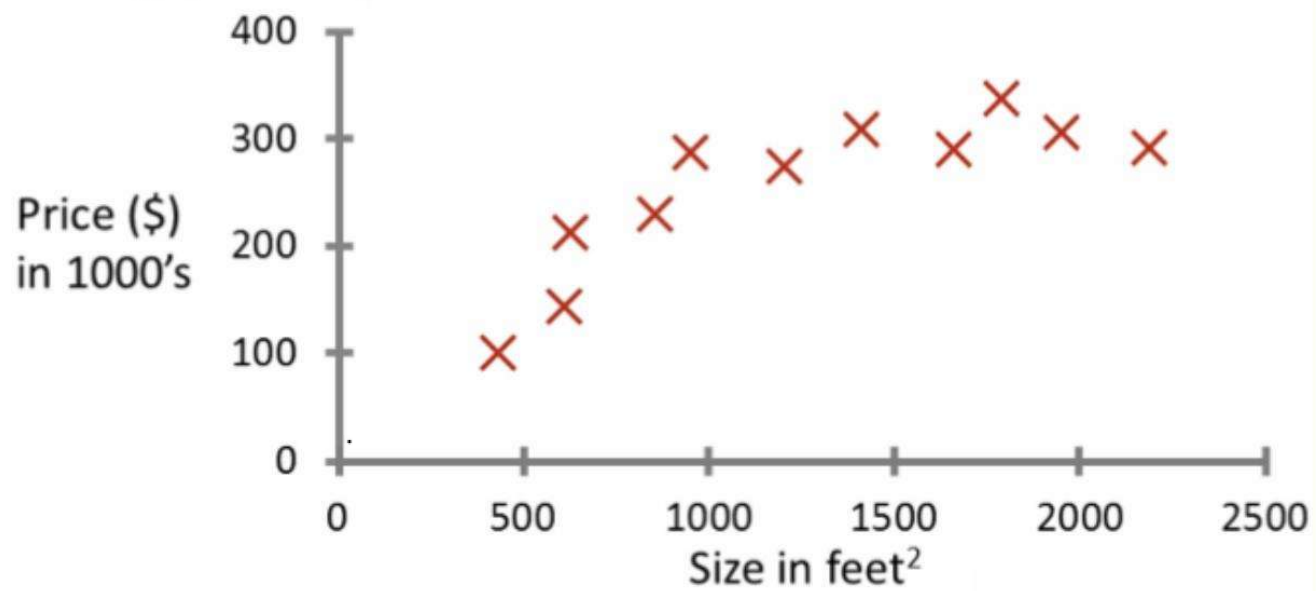
- **Supervised learning** is a type of machine learning where algorithms learn from labeled data (input-output pairs) to find patterns, enabling them to make predictions or decisions on new, unseen data,

- In supervised learning:
- We have **input + correct output**
- The goal is to **predict the output for new (future) inputs**
- So the model learns a function:

$$X \rightarrow f(\) \rightarrow Y$$

- Example:
- X = house size, bedrooms
- Y = price
- The trained model **predicts price** for a new house
- Therefore supervised ML produces a **prediction model**

Housing price prediction.



Supervised → Two subtypes

1. Classification (predict *categories*)

- **Email spam detection**

Input: email text

Output: *spam / not spam*

- **Medical diagnosis**

Input: X-ray

Output: *healthy / pneumonia*

- **Credit card fraud detection**

Input: transaction details

Output: *fraud / not fraud*

- Regression (predicts a real value (scalar, vector))
- **House price prediction**
Input: size, bedrooms, location
Output: €250,000
- **Stock price estimation**
Input: financial trends
Output: *price next hour/day*
- **Predicting time for delivery**
Input: distance, traffic
Output: *minutes*

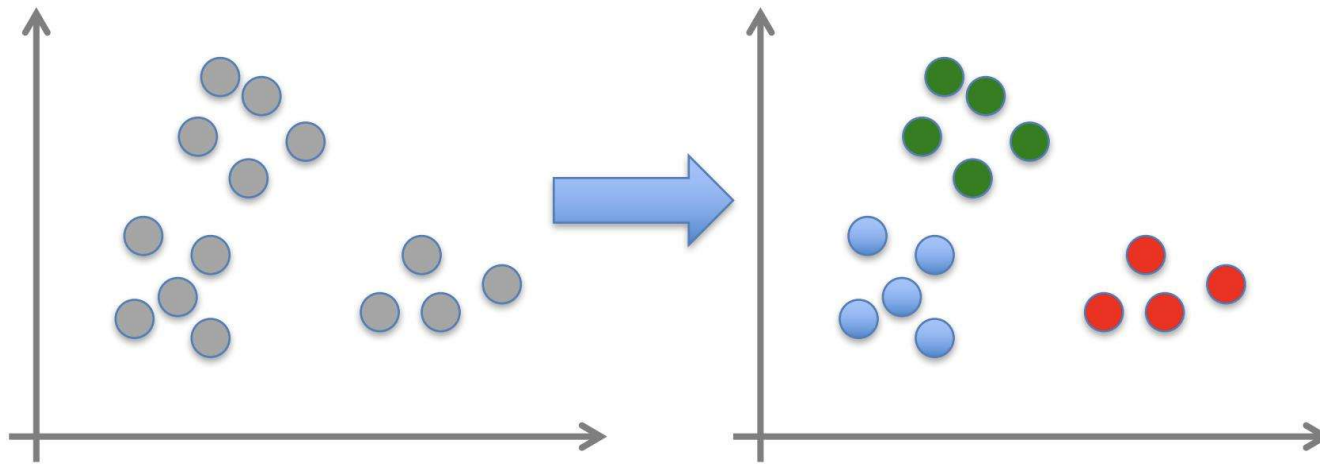
Unsupervised

- **Unsupervised learning is when the model learns patterns from data that has no labels.**
- No “correct answer” is given
- The model tries to discover structure in the data

- In unsupervised learning:
- We have only inputs, no labels
- The goal is NOT prediction
- The goal is to understand the structure of the data
- There is no Y to predict
- Therefore unsupervised ML produces a **data model**
(i.e., a representation of how the data is organized internally)

- Examples: clustering algorithms, association
- Customer segmentation- e.g. companies group customers together based on similarities like, age or location or spending habits
- Finds hidden patterns or similarities
- Association is where the algorithm looks for relationships between variables in the data.
- E.g. companies want to know which items are often bought together.

Example-clustering



Exercise

- **What kind of models would each of these tasks use?**
- **► Predict tomorrow's rainfall**, given the date and the last 10 days of rainfall data
- **► Determine from a photo of a face** if the user is who she claims to be
- **► Estimate the probability of 10 possible diagnoses**, given some patient data, test results
- **► Cluster customers** into 22 different groups with similar buying habits
- **► Estimate the risk (probability)** of an auto accident at a location given the hour and weather

The Machine Learning Workflow

- Collect data
- Clean & prepare data
- Split data (training set/test set)
- Choose hypothesis/model
- Train the model
- Evaluate on test set
- Improve (features, parameters)

Key ML Terms

- **Training set:** The portion of the data used to teach the model patterns.
- **Test set:** A separate portion of the data used to see how well the trained model performs on unseen examples.
- **Features (x):** The input variables or measurable properties given to the model (e.g., size, age, color).
- **Target/label (y):** The correct answer the model is trying to predict (e.g., price, spam/not spam).

Key ML Terms

- **Hypothesis (model):** The learned function that maps inputs (x) to predicted outputs (y).
- **Prediction:** What the trained model outputs when given new data.
- **Error / accuracy:** A number that tells how well the model performed — how close predictions were to the correct answers.
- **Learning algorithm:** The method or recipe the model uses to learn from the training data (e.g., kNN, SVM, linear regression).

A Simple ML Model Example

- Problem 1: Predict whether a student passes based on hours studied.
- Problem 2: Predict student score based on hours studied.

Training vs Testing

- Never evaluate your model on the same data you trained it with.
- train/test split matters
- Overfitting

Common misunderstandings to prevent early

- ML is not magic → it needs data
- More data $\neq$ always better → needs quality
- Models don't think → they detect patterns